

March 15, 2025

Networking and Information Technology Research and Development National Coordination Office

National Science Foundation
490 L'Enfant Plaza SW, Suite 8001
Washington, DC 20024

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

To whom it may concern:

After a decade of research and analysis on immigration, workforce, and technology issues, FWD.us understands that immigration is one of the United States' greatest competitive advantages. We are especially attentive to the importance of policies to win the global race for talent by attracting and retaining individuals with advanced education and specialized training in critical industries,¹ including artificial intelligence. We look closely and critically at immigration policies that affect individuals with education and training in science, technology, engineering, and math (STEM) fields, including international students² who earn their degrees at U.S. colleges and universities.

In this spirit, we submit these comments in response to the Request for Information (RFI) titled, "Request for Information on the Development of an Artificial Intelligence (AI) Action Plan."

A. AI Action Plan must address pressing talent shortages and skills gaps

On January 23, 2025, President Trump signed Executive Order 14171, "Removing Barriers to American Leadership in Artificial Intelligence," establishing the administration's goal to "solidify our position as the global leader in AI and secure a brighter future for all Americans."

To achieve this goal, the United States will need a plan to attract and retain more individuals with highly specialized education, skills, and experience to contribute to AI ventures here.

¹ FWD.us, "Strengthening America's Competitiveness and Security by Welcoming More Immigrants With STEM Skills," April 2023, <https://www.fwd.us/news/stem-immigrants>.

² FWD.us, "Retaining U.S. International Student Graduates Could Help the U.S. Win the Global Talent Race," February 2022, <https://www.fwd.us/news/us-international-students>; FWD.us, "Advanced International STEM Students Increasingly Anticipate Shorter-Term Work Stays After Graduation. The U.S. Can Do More To Retain Them," September 2024, <https://www.fwd.us/news/stem-international-students>.

One significant but solvable challenge to America's AI competitiveness is the fact that there are not enough qualified individuals in the U.S. workforce to meet the needs of current, much less future, AI ventures. A 2019 analysis by the Center for Security and Emerging Technologies (CSET) at Georgetown University found that "there is broad consensus in the field that talent shortages are substantial,"³ and a more recent Reuters report estimates current AI talent shortages at 50%.⁴ A 2023 report from global consulting firm McKinsey found that many employers are still struggling to fill AI-related positions.⁵ As other sectors incorporate AI more deeply, the AI skills gap will challenge their future growth, as well.⁶

Addressing these talent gaps will certainly involve optimizing the U.S. workforce to fill roles in these industries, including increasing the number of domestic students who earn advanced degrees in STEM fields and train for jobs in relevant sectors, and "reskilling" or "upskilling" to support current workers adapting to new and changing fields.

However, an AI talent plan must also acknowledge, account for, and seek to maximize the contributions made by foreign-born experts, researchers, and workers with relevant STEM training and experience in AI and related sectors.

B. Foreign-born STEM experts are a critical component of our AI workforce

Immigrants make up a significant share of the STEM workforce in the U.S. The National Science Foundation's (NSF) 2024 Science and Engineering Indicators report found that "in 2021, foreign-born workers (regardless of citizenship status) accounted for 19% of the STEM workforce. Foreign-born workers accounted for 19% of workers in S&E occupations at the bachelor's degree level, 37% at the master's degree level, and 43% at the doctorate level."⁷

International students who graduate from U.S. colleges and universities are a significant part of the foreign-born STEM workforce. Since 2008, about 44% of all advanced STEM degrees awarded by U.S. colleges and universities have gone to international graduates, and about 46% of all advanced degrees awarded to international graduates were in STEM fields.⁸ These trends hold true for AI, where "the majority of workers in AI-related jobs and students in AI-related graduate programs are not originally from the United States."⁹

³ Remco Zwetsloot, "Strengthening the U.S. AI Workforce," Center for Security and Emerging Technologies (CSET), September 2019, <https://cset.georgetown.edu/publication/strengthening-the-u-s-ai-workforce>.

⁴ Mark Dangelo, "Needed AI skills facing unknown regulations and advancements," Reuters,

⁵ McKinsey, "The state of AI in 2023: Generative AI's breakout year," August 2023, <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year#talent>.

⁶ SecurityIntelligence, "ISC2 Cybersecurity Workforce Study: Shortage of AI skilled workers," January 2025, <https://securityintelligence.com/articles/isc2-cybersecurity-workforce-study-shortage-ai-skilled-workers/>.

⁷ Steven Deitz and Christina Freyman, "The State of U.S. Science and Engineering 2024," National Science Foundation, March 2024, <https://nces.nsf.gov/pubs/nsb20243>.

⁸ FWD.us, "High-Skilled Immigration: 5 Things to Know," February 2025, <https://www.fwd.us/news/high-skilled-immigration-5-things-to-know>.

⁹ Zwetsloot, "Strengthening the U.S. AI Workforce."

It is clear that the current AI workforce relies significantly on the contributions of immigrants, and that immigrants will continue to be an important part of the AI workforce into the future, especially in addressing immediate workforce needs. A recent Center for Strategic and International Studies analysis explains that “while training and educating STEM talent domestically among elementary school and high school students will address future workforce needs, it is more of a generational strategy—one that does not solve the acute, near-term shortages that high-tech industries and U.S. federal S&E-related agencies struggle with.”

In the short- and long-term, immigration is a powerful complement to the U.S.-born workforce and a potentially key competitive advantage for the U.S. in the global AI race. Commenting on the NSF’s findings, a National Science Board member concurred, noting that, “Going forward, it’s vital that we cultivate the diversity of our domestic STEM talent while strengthening our capacity to attract and retain international students and professionals.”¹⁰

C. Increasing retention of global talent will strengthen America’s hand in the global AI race

For decades, the U.S. has been the premier destination for individuals seeking the best higher education and career opportunities in STEM fields. The fact that the world’s best and brightest minds want to come *here* above anywhere else is a tremendous competitive advantage. But we cannot take for granted that international STEM experts will *continue* to see the U.S. as the top choice to earn their education and contribute their skills, particularly in the face of aggressive recruiting challenges by other countries.

CSET explains that “other countries, most notably China, are actively trying to transfer and absorb AI knowledge and technology from the United States for their industries and militaries.”¹¹ This includes recruiting current AI experts and talent away from the U.S. and attracting future talent by recruiting international students to study in their countries instead of the U.S.

Their efforts are helped by restrictive immigration policies in the U.S. that are diminishing international students’ ability and interest in coming to the U.S. to study or remaining in the U.S. to work after graduation.¹² These restrictions are also hampering U.S. companies’ ability to hire employees to fill positions here, leading them to move operations and hiring outside the U.S.¹³

However, the U.S. can defeat these challenges and strengthen its competitive advantage by taking steps to increase the attraction and retention of global talent. Retaining talent is a twofold victory for the U.S.; not only do we benefit from their contributions (and in the case of

¹⁰ “New report shows the importance of foreign-born talent to the U.S. STEM workforce,” National Science Foundation, May 2024, https://www.nsf.gov/nsb/news/news_summ.jsp?cntn_id=309769.

¹¹ Zwetsloot, “Strengthening the U.S. AI Workforce.”

¹² FWD.us, “Advanced International STEM Students Increasingly Anticipate Shorter-Term Work Stays After Graduation. The U.S. Can Do More To Retain Them.”

¹³ Tina Huang and Zachary Arnold, “Immigration Policy and the Global Competition for AI Talent,” CSET, June 2020, <https://cset.georgetown.edu/publication/immigration-policy-and-the-global-competition-for-ai-talent>.

international graduates, reap the rewards of our investment in their education), but we also deny our competitors access to their innovation and productivity. In industries that are defined by intense races to the next breakthrough and incredibly limited access to talent, every additional person we can recruit to our team adds tremendous value and advantage to our side.

President Trump¹⁴ and his close advisors¹⁵ have commented publicly about the importance of modernizing policies to attract and retain top global talent. Leaders in the AI space agree; for example, the CEO of the Allen Institute for Artificial Intelligence wrote in 2019 that the Trump administration's AI efforts should "include a special visa program for AI students and experts to help us win this race for the sake of both economic vitality and national security."¹⁶

As the Trump administration prepares its next set of AI priorities and orders, it should include concrete measures to increase the attraction and retention of global AI talent, particularly international graduates of U.S. colleges and universities.

D. Opportunity to retain more international students graduating from U.S. schools

Fortunately, there are already significant numbers of international students graduating from U.S. colleges and universities each year with degrees relevant to AI.

Based on graduation statistics in recent years, we anticipate that some 25,000 international students likely graduate with AI-related advanced degrees in the U.S. each year, including in computer science and computer engineering, according to FWD.us analysis of National Center of Education Statistics.¹⁷ And likely as many as 19,000 of those students would like to remain in the U.S. for some time after graduation, based on a FWD.us survey of prospective international students seeking advanced degrees.¹⁸

More broadly, FWD.us analysis projects that 100,000 additional international student graduates each year would stay and work permanently in the U.S. if a pathway existed, adding up to \$233 billion to the U.S. economy and reducing STEM-related talent shortages by about a quarter.¹⁹

¹⁴ Alexandra Ulmer and Gram Slattery, "Trump says foreigners who graduate from US colleges should get green cards," Reuters, June 2024, <https://www.reuters.com/world/us/trump-says-foreign-college-graduates-should-automatically-get-green-cards-2024-06-20>.

¹⁵ Chris Pandolfo, "Elon Musk combats anti-immigration sentiment in posts decrying 'dire shortage' of tech talent," Fox Business, December 2024, <https://www.foxbusiness.com/politics/elon-musk-combats-anti-immigration-sentiment-posts-decrying-dire-shortage-tech-talent>.

¹⁶ Oren Etzioni, "What Trump's Executive Order on AI Is Missing," WIRED, September 2019, <https://www.wired.com/story/what-trumps-executive-order-on-ai-is-missing>.

¹⁷ FWD.us, "Modernizing Schedule A for Critical and Emerging Industries," January 2024, <https://www.fwd.us/news/schedule-a-rfi#ai>.

¹⁸ FWD.us, "Retaining U.S. International Student Graduates Could Help the U.S. Win the Global Talent Race."

¹⁹ Ibid.

However, the majority of international student graduates, including significant shares of advanced degree recipients, do not remain in the U.S. after graduation, in large part because of a lack of pathways available to them.²⁰ While stay rates for STEM PhD graduates are much better,²¹ they still face a long, complicated, and expensive bureaucratic road to secure the kind of permanent status that provides certainty about their future and allows them to work on more restricted government-funded projects in these specialized fields that typically require at least a green card.

E. Policy Recommendations to Increase Retention and Streamline Processes

FWD.us has published numerous resources outlining policy recommendations to streamline outdated processes in the legal immigration system and increase the retention of international student graduates.²² We urge the administration to include these policy priorities in its AI plan:

- **Preserve and improve legal immigration pathways for international student graduates, including Optional Practical Training (OPT), H-1B, and O-1.**

These programs are important for the attraction *and* retention of global talent, including students. The availability of post-graduation work opportunities is a significant factor shaping prospective students' decision to study in the United States.²³ The Optional Practical Training (OPT) program gives students hands-on experience in their field of study as well as opportunities to build career skills and establish relationships with future employers.²⁴ After graduating, skills-based pathways like the H-1B²⁵ and O-1 visas are the primary options for international graduates to remain and work in the U.S. until they can secure permanent residency.

Recent administrative efforts to modernize processes, clarify guidance, and expand access to these programs have been very successful in strengthening retention in the face of declining enrollment and student interest in longer-term stays.²⁶ This administration should continue and increase outreach efforts to make sure students and employers are aware of these available pathways, and continue to regularly update guidance for the public and adjudicators to provide clarity about who can qualify and what evidence is most effective. These programs are already in place—we should be getting the most out of them. The administration should also clearly commit to preserving and

²⁰ Connor O'Brien, "Most International Graduates of American Universities Ultimately Leave the U.S.," Economic Innovation Group, June 2024, <https://eig.org/immigrant-retention-estimates>.

²¹ Jack Corrigan, James Dunham, and Remco Zwetsloot, "The Long-Term Stay Rates of International STEM PhD Graduates," CSET, April 2022, <https://cset.georgetown.edu/publication/the-long-term-stay-rates-of-international-stem-phd-graduates>.

²² FWD.us, "Retaining International Student Graduates: 5 Policy Proposals for Congress," October 2024, <https://www.fwd.us/news/retaining-international-student-graduates>.

²³ Shorelight, "The Essential Role of Practical Training in U.S. Higher Education and the Innovation Economy," May 2024, <https://shorelight.com/student-stories/the-essential-role-of-practical-training-in-u-s-higher-education>.

²⁴ FWD.us, "What is OPT? Optional Practical Training Policy Brief," October 2024, <https://www.fwd.us/news/what-is-opt>.

²⁵ FWD.us, "The H-1B Visa Program: Policy Brief," January 2025, <https://www.fwd.us/news/h1b-visa-program>.

²⁶ FWD.us, "Advanced International STEM Students Increasingly Anticipate Shorter-Term Work Stays After Graduation. The U.S. Can Do More To Retain Them," September 2024, <https://www.fwd.us/news/stem-international-students>.

strengthening the OPT program, to maximize its effectiveness and to communicate to prospective students that it will remain available should they choose to study in the U.S.

- **Streamline the legal immigration process through the use of existing efficiency mechanisms like Schedule A.**

Whereas the previous administration failed to follow through on the thousands of public comments expressing support for restoring and modernizing the Schedule A process, this administration has an opportunity to lead in streamlining legal immigration processes, saving time and taxpayer dollars while cutting through bureaucratic red tape. This simple policy change was recommended by the bipartisan House Select Committee on the Strategic Competition Between the United States and the Chinese Communist Party.²⁷

As we described in our own response,²⁸ establishing an objectively measurable, self-executing filter, so that intensive processes like permanent labor certification (PERM) are only required when necessary to protect U.S. workers, would strengthen America's hand in the global competition for talent by providing more predictability for highly educated and skilled STEM experts considering job opportunities in the U.S., and for the companies seeking to hire them.²⁹

FWD.us analyzed PERM filings over fiscal years 2021, 2022, and 2023 to identify any trends that might be evident in the data from employers that file the largest numbers of PERM applications for advanced degree holders.³⁰ Looking at the employers who filed at least 500 PERM requests over the three most recent fiscal years for positions where a Master's or above was the minimum educational requirement, there are a number of occupations for which PERM certifications were *granted* more than 99% of the time. And nine out of the top ten of these occupations are STEM jobs. Occupations like these would be strong candidates for consideration in an updated Schedule A list.

By reducing the PERM workload for DOL, this expedited process would save taxpayer dollars as well. According to FWD.us analysis, an annual average of about 70,000 PERM applications were filed for workers in occupations that we classified as STEM over the last three years. If

²⁷ The Select Committee on the Strategic Competition Between the United States and the Chinese Communist Party, "Reset, Prevent, Build: A Strategy to Win America's Economic Competition with the Chinese Communist Party," <https://selectcommitteeontheccp.house.gov/sites/evo-subsites/selectcommitteeontheccp.house.gov/files/evo-media-document/reset-prevent-build-scc-report.pdf>.

²⁸ FWD.us, "Modernizing Schedule A for Critical and Emerging Industries," January 2024, <https://www.fwd.us/news/schedule-a-rfi>.

²⁹ While we are not proposing a specific methodology for creating or implementing such a filter, we encourage the Department to seriously consider proposals put forward by experts such as the Institute for Progress (see for example Institute for Progress, "Help Wanted: Modernizing the Schedule A Shortage Occupation List," December 2023, <https://ifp.org/schedule-a>).

³⁰ FWD.us analysis of Department of Labor PERM Disclosure Data (accessed at <https://www.dol.gov/agencies/eta/foreign-labor/performance>) for fiscal years 2021, 2022, and 2023. Data are filtered to show PERM certification filings for occupations requiring at least a Master's degree, and by employers who submitted at least 500 certification filings for such occupations over fiscal years 2021-2023.

these STEM occupations had been included on Schedule A, this would have saved nearly \$6.2 million on average annually to process the applications.³¹

- **Address the impacts of green card backlogs on long-term workers and their families living in the U.S.**

Outdated laws governing the issuance of green cards to employment-based immigrants have created a backlog of individuals who qualify for permanent residency but may have to wait decades until they actually get their green cards.³² During this time, their ability to change jobs or employers is heavily restricted, as is their spouses' ability to work to support themselves, their families, and their communities.³³ While waiting in backlogs, their children are at risk of aging out of their dependent status and being forced to leave the country.³⁴ Research shows that green card backlogs motivate highly skilled immigrants, many who have already lived in the U.S. for a long time, to consider leaving for other countries with more family-friendly policies.³⁵

There are steps the administration can take to relieve families of the negative effects of the green card backlog. The administration could clarify eligibility for case-by-case relief for children of backlogged immigrants, and eligibility for work permits under "compelling circumstances." The H-4 EAD program, which provides work authorization for certain spouses of backlogged H-1B visa holders seeking green cards, has been a resounding success and should be preserved. In addition, the administration should explore available actions to expedite access to green cards for people in the backlog, such as administrative recapture of previously wasted employment-based green cards³⁶ and potential changes to the visa bulletin.³⁷

³¹ Analysis is based on Department of Labor data (<https://www.dol.gov/agencies/eta/foreign-labor/performance>) reporting the number of applications processed each year. The cost-saving analysis was produced by estimating the share of DOL's Office of Foreign Labor Certification annual budget allocation (<https://www.dol.gov/general/budget>) that went towards processing PERM applications, and applying that cost to the share of PERM determinations that were for STEM and Healthcare occupations. PERM filings were sorted by their Standard Occupational Classification (SOC) according to the Bureau of Labor Statistics; for STEM, we included all occupations classified under SOC 15 (Computer and Mathematical Occupations), SOC 17-2000 (Engineers), and SOC 19 (Life, Physical, and Social Science Occupations); for healthcare, we included SOC 29 (Healthcare Practitioners and Technical Occupations) and SOC 31 (Healthcare Support Occupations). The calculations were based on annual averages for fiscal 2021, 2022, and 2023.

³² FWD.us, "Per-Country Cap Reform: Priority Bill Spotlight," December 2023, <https://www.fwd.us/news/per-country-cap-reform-priority-bill-spotlight>.

³³ FWD.us, "H-4 EAD is Smart Policy that Should Be Preserved," November 2024, <https://www.fwd.us/news/h4ead>.

³⁴ FWD.us, "Children of immigrants in green card backlogs face uncertain futures in the U.S.," January 2024, <https://www.fwd.us/news/children-of-immigrants-in-green-card-backlogs-face-uncertain-futures-in-the-u-s>.

³⁵ Pooja B. Vijayakumar and Christopher J. L. Cunningham "An Indentured Servant: The Impact of Green Card Waiting Time On the Life of Highly Skilled Indian Immigrants In the United States of America," Industrial and Organizational Psychology Translational Research and Working Papers, January 2019, <https://scholar.utc.edu/cgi/viewcontent.cgi?article=1002&context=iopsy>.

³⁶ FWD.us, "Green card recapture and reform would reduce immigration backlogs," January 2025, <https://www.fwd.us/news/green-card-recapture>.

³⁷ Stuart Anderson, "Changing Visa Bulletin Could Save Immigrants And H-1B Visa Holders," Forbes, September 2023, <https://www.forbes.com/sites/stuartanderson/2023/09/06/changing-visa-bulletin-could-save-immigrants-and-h-1b-visa-holders>.

- **Engage Congress on passing legislation to modernize the legal immigration system and establish better options for international graduates.**

Ultimately, only Congress can act to modernize our immigration laws and establish reliable pathways to attract and retain global talent, including international graduates and entrepreneurs. Despite the pressing need and plenty of good proposals,³⁸ Congress has not mustered the political will to pass bipartisan legislation on this issue.

In addition to administrative steps to streamline processes and improve processes for AI talent, we also urge this administration to directly engage leaders in Congress to propose and pass legislation to modernize our legal immigration system, to make it faster, easier, and more cost-effective for global talent to come and stay in the U.S. and contribute their talents here, particularly people with education and skills relevant to our AI efforts.

F. Conclusion

The global race for talent will shape America's ability to lead in any number of critical areas, including the AI race. We have built the world's best system for training and educating top talent; now we must make sure we can capitalize on those investments. We strongly encourage you to incorporate these proposed actions and policies to strengthen recruitment and retention of global talent, preserving one of the United States' greatest competitive advantages into the future.

Respectfully submitted,

Todd Schulte

President

On behalf of FWD.us

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³⁸ FWD.us, "Retaining International Student Graduates: 5 Policy Proposals for Congress."